

SIDDHARAMAYYA MATHAPATI

SENIOR AI ENGINEER

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PORTFOLIO | GitHub | LINKEDIN

EDUCATION

ACHARYA INSTITUTE OF TECHNOLOGY
MASTER'S COMPUTER APPLICATIONS
CGPA: 7.2

2018 - 2020
Bangalore

B K COLLEGE
BACHELOR COMPUTER APPLICATION
CGPA: 6.8

2015 - 2018
Chikodi

EXPERIENCE

MAVERIC SYSTEMS LIMITED | SENIOR ARTIFICIAL INTELLIGENCE ENGINEER
June 2025 - Present

Bangalore

Built a production-grade multi-agent AML investigation platform using LangGraph and AutoGen, implementing an 8-node sequential pipeline that ingests real-time Kafka alert streams, performs behavioral analysis across 7 risk dimensions, and auto-generates SAR reports aligned with L1 AML investigation frameworks.

Integrated AWS Bedrock (Mistral, Titan Embeddings) with MongoDB Atlas Vector Store and LangSmith distributed tracing, implementing PII redaction, per-agent latency monitoring, and schema-validated JSON outputs to ensure auditability and compliance in financial decisioning systems.

Designed and implemented PRISM, an end-to-end AI pipeline converting regulatory PDFs into executable DMN/DRL rules, deployed via Apache KIE/Drools; built a 7-stage asynchronous LLM workflow using Gemini 2.5 Flash, with OpenTelemetry observability and four-eyes governance (90.6% completion across 18 validation gates).

Architected CodeLens, a microservices-based AI code analysis platform with 7 independent FastAPI agent services (Ingestion, Parsing, Explainer, Quality, Dependency, Refactor, Coordinator), integrating Azure DevOps, GitHub, and SQL Server, using Tree-sitter AST parsing and Ollama-hosted LLMs.

Developed a spec-compliant MCP (Model Context Protocol) server, exposing 10+ banking tools and 11 fraud intelligence resources over HTTP, WebSocket, and STDIO transports, integrated with AutoGen multi-agent orchestration using Qwen2.5 and Gemma models via Ollama.

Built a real-time IT Operations Command Center UI using React 19, TypeScript, and Tailwind CSS, leveraging Server-Sent Events (SSE) for live alert streaming, D3-Geo visualizations, auto-healing dashboards, and sub-second latency updates.

Implemented a comprehensive LLM observability framework using LangSmith, including traceable decorators, A/B prompt experimentation with statistical evaluation, human feedback annotation pipelines, and prompt versioning via LangSmith Hub.

Designed an AI-powered loan delinquency prediction system with 0-999 risk scoring, delivering context-aware repayment strategies, scenario classification (temporary vs chronic financial stress), and multi-channel personalized communication workflows, targeting NPA reduction and credit risk optimization.

- Fine-tuned LLMs using QLoRA and LoRA, optimizing model efficiency while reducing memory and computational overhead.
- Prepared, cleaned, and curated datasets for fine-tuning LLMs, ensuring high-quality data for domain-specific adaptation.
- Developed and optimized attention mechanisms to enhance transformer-based model performance for NLP tasks.
- Built and optimized training loops for large-scale AI models, ensuring efficient convergence and distributed training across GPUs.
- Designed and deployed Retrieval-Augmented Generation (RAG) applications, integrating vector databases and hybrid search for accurate responses.
- Developed domain-specific LLMs, fine-tuned for industries such as finance, healthcare, and cybersecurity to improve contextual accuracy.
- Implemented real-time AI inference pipelines using TensorRT and ONNX for low-latency model deployments.
- Leveraged reinforcement learning (RLHF) to fine-tune LLMs, improving alignment with human feedback. Integrated vector databases (FAISS, ChromaDB) for efficient document retrieval in large-scale knowledge management systems.
- Deployed AI workloads on cloud platforms (AWS, GCP, Azure), optimizing costs and ensuring high availability.

X- CENCIA TECHNOLOGY SOLUTION LIMITED IND | PROJECT INTERN

Bengaluru, KA |

Feb 2020-April 2021

- Trained and deployed transformer-based NLP models for real-time text generation and summarization.
- Designed an automated data pipeline for processing sensor data from ESP32 and Raspberry Pi-based devices.
- Developed a smart school bus tracking system using RFID, GPS, and IoT, ensuring real-time location monitoring of students.
- Developed a real-time AI-based security system for cars using IoT, facial recognition, and RFID authentication.

PROGRAMMING LANGUAGES	Python, Bash/Shell Scripting
LIBRARIES/Frameworks	JavaScript, PySpark, Langchain, Django, Large Language Models (LLM), Artificial Intelligence (AI), Machine Learning and Operations(MLOPS), Retrieval- Augmented Generation (RAG), Natural Language Processing (NLP), Google Cloud Platform, Amazon Web Services, TensorFlow
Tools / Platforms	Git, Vs Code, MLFLOW, LangGraph, Kubernetes, AWS SageMaker, HuggingFace,
DATABASES	MySQL, PostgreSQL, VectorDB, ChromaDB, FAISS

PROJECTS / OPEN-SOURCE

LINKEDIN-COMMENT-POSTER-WITH-AI | [LINK](#)

Python

- A Python automation tool using Selenium and BeautifulSoup to extract user activities—posts, likes, shares, and comments—while handling dynamic content, errors, and ensuring ethical scraping for analysis.

RECRUITMENT_HELPER_AI_WITH_LLAMA | [LINK](#)

Jupyter Notebook, Python

- An AI-powered tool that scrapes job info from career pages and generates personalized cold emails. Streamlines recruitment outreach by eliminating manual effort and boosting efficiency.

STOCK-PRICE-PREDICTION-USING-LLMS | [LINK](#)

Python,

- An intelligent assistant that uses LLMs to analyze stock trends, predict prices, and summarize financial news using data from Yahoo Finance and NewsAPI. Designed to support smarter investment decisions.

CREDIT-RISK-ANALYSIS-WITH-CATBOOSTCLASSIFIER | [LINK](#)

Python

- A machine learning-based solution that predicts the likelihood of credit card defaults. Designed to help financial institutions assess credit risk, reduce defaults, and enhance profitability through automated, data-driven decision-making.

MEDICAL-ASSISTANT-AI | [LINK](#)

Python

- This project is a web application that utilizes Google Generative AI to assist medical professionals by analyzing medical images and providing diagnostic insights. The application allows users to upload images and receive analytical feedback through a chat interface.

SIMPLE_SQL_WITH_GOOGLE_GEMINI_AI | [LINK](#)

Jupyter Notebook, Python

- Integrated Google Gemini LLM with MySQL using LangChain to convert natural language queries into accurate SQL. Utilizes semantic similarity and few-shot learning for precise, context-aware data retrieval.

STUDENT_STUDY_HELPER_AI_WITH_GOOGLE_GEMINI | [LINK](#) Python

- This Project provides a chatbot built using LangChain, Gemini LLM, and PDF processing to answer questions based on uploaded PDF documents. The chatbot is designed to help students by answering their queries from the context of the uploaded files.

CERTIFICATIONS

- LLM Engineering: Master AI, Large Language Models & Agents - UDEMY
- MLOps Bootcamp: Mastering AI Operations for Success Udemmy - UDEMY
- Python for Data Analysis & Visualization - UDEMY
- Deep Learning Masterclass with TensorFlow 2 - UDEMY